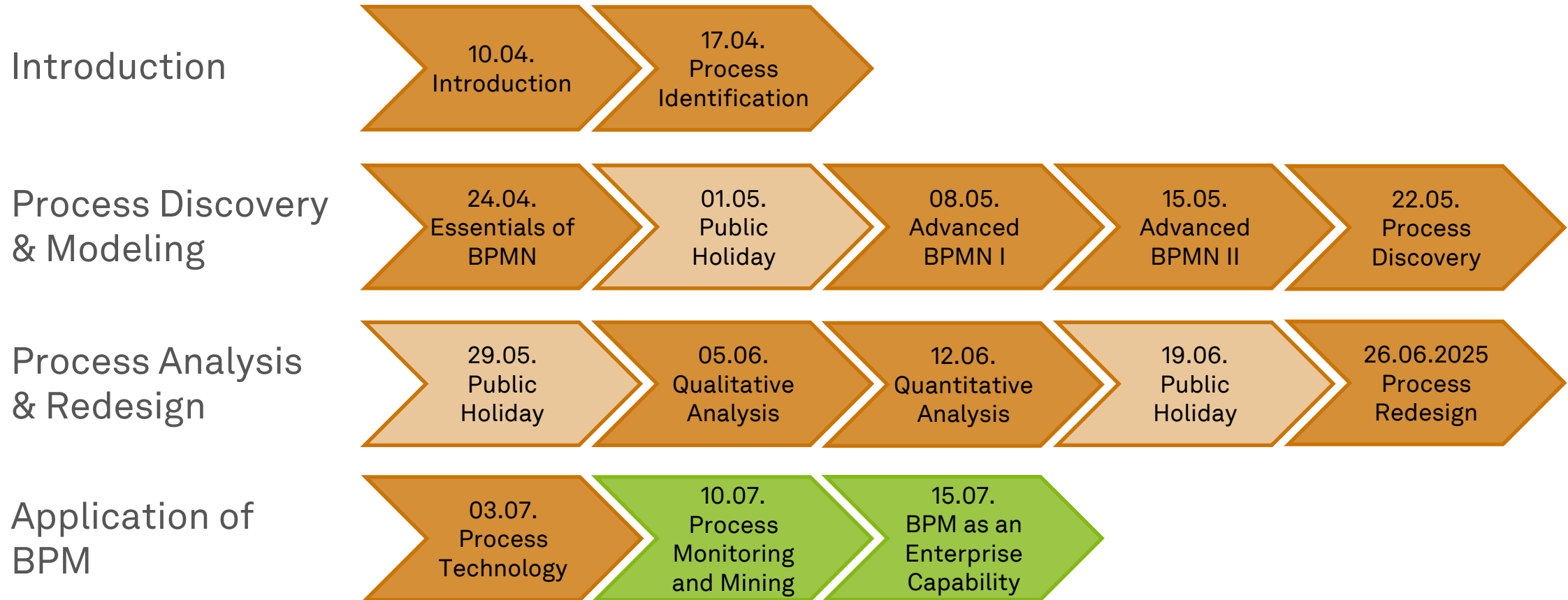


Business Process Management

Process Monitoring and Process Mining

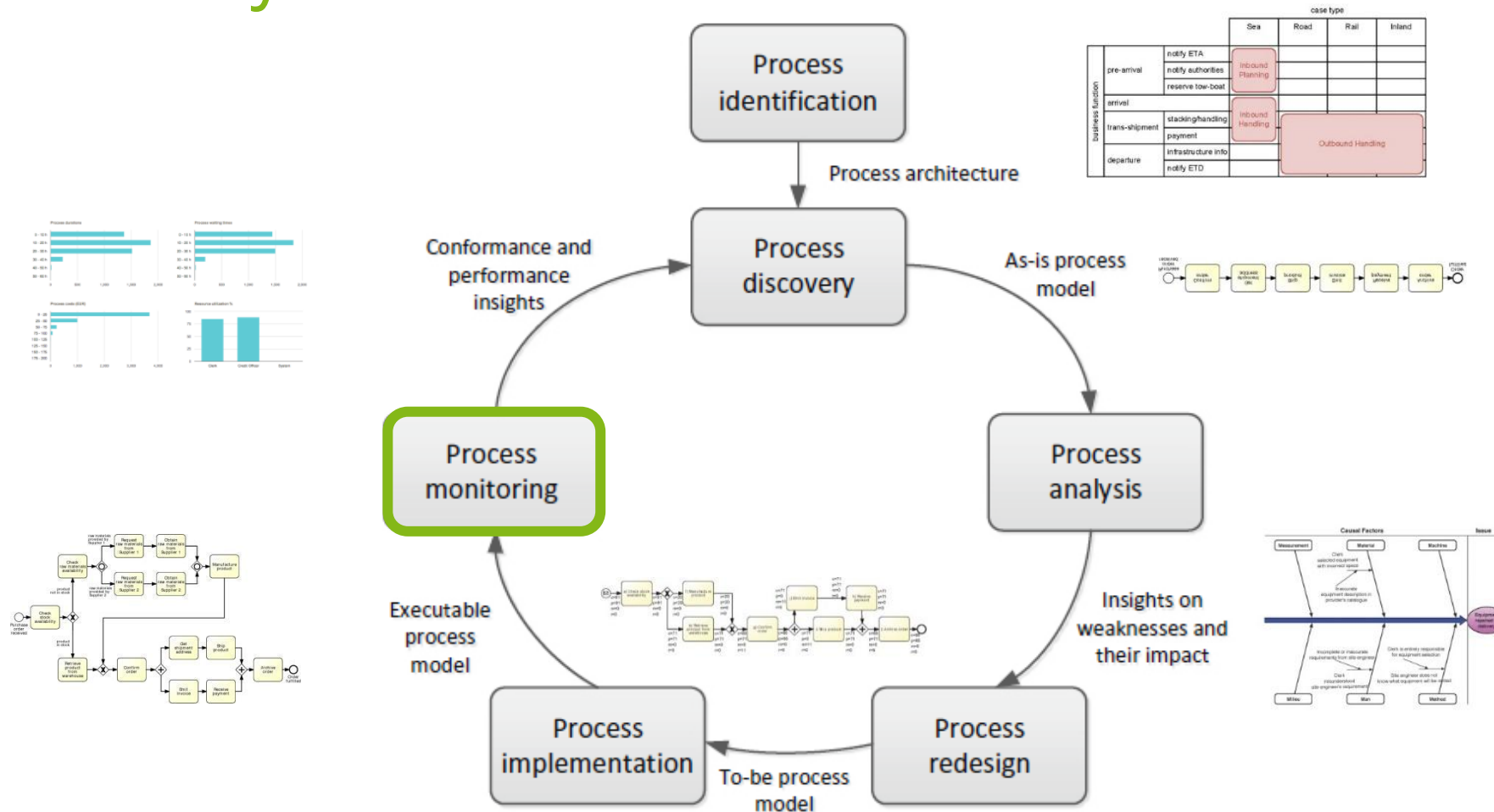


Roadmap



Learning Goals of Today

- Understand process monitoring with process dashboards
- Understand automated process discovery
- Understand process conformance checking
- Understand variants analysis and process performance mining



Process Monitoring (1)

- Use the data generated by the execution of a business process
 - **Event record** captures state change in execution of **work items**
 - Collection of time-stamped event records is an **event log**
- Extract insights about the actual **model** and **performance** of the process
- Verify its **conformance** with respect to norms, policies, or regulations
- Analysis can be done
 - Post-mortem (**offline**) or on-the-fly (**online**)
- Techniques are
 - **Statistics**-based (dashboards) or **model**-based (process mining)

Process Monitoring (2)

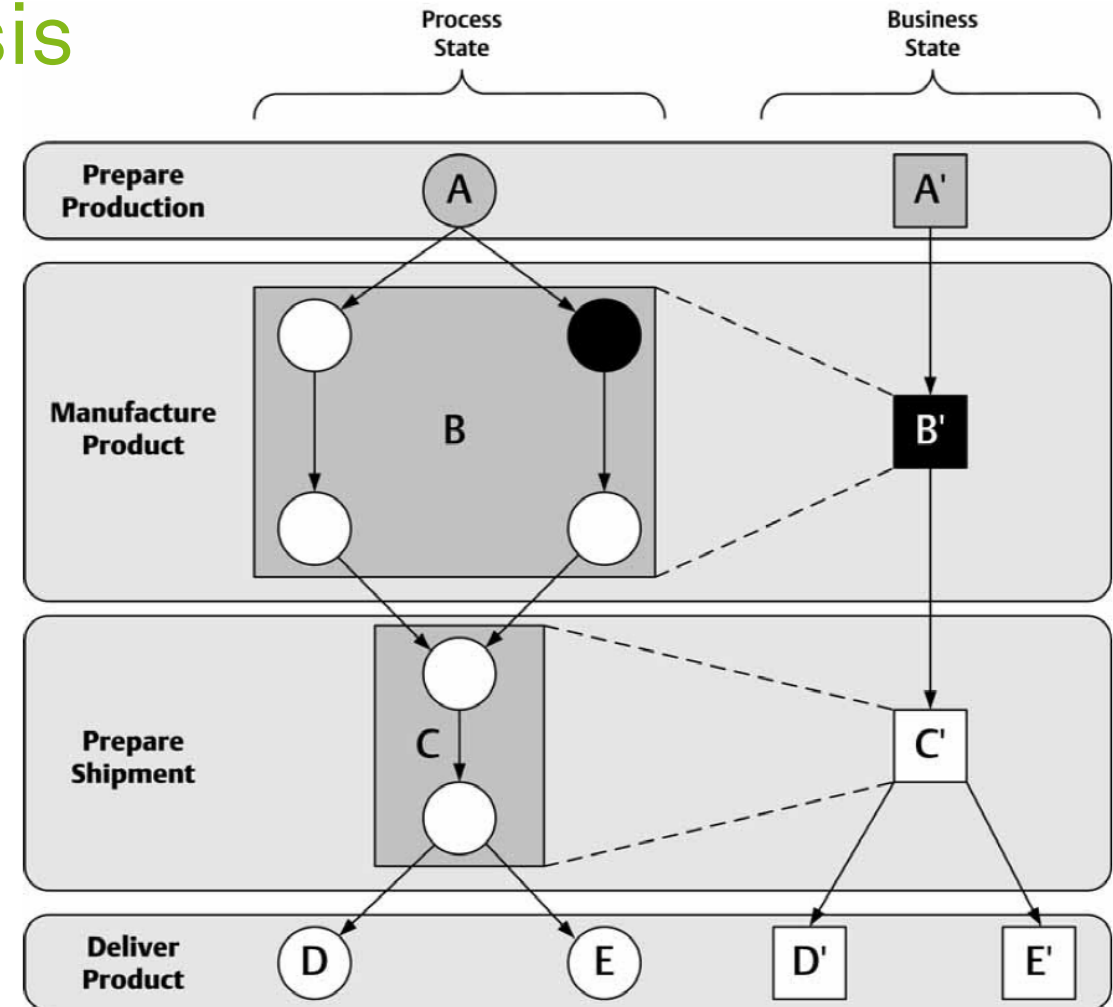
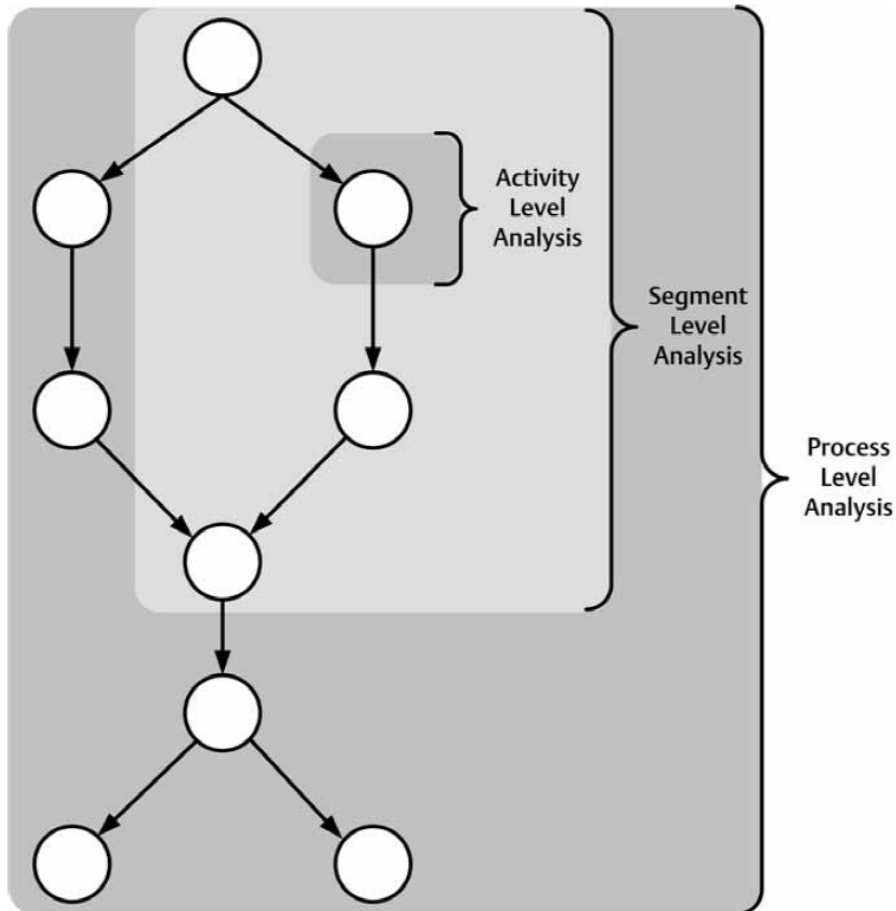
- **Offline Process Monitoring**

- Covers **historic** process executions
- Provides a picture of
 - the **performance** of the process
 - the **reasons** for
 - poor performance or
 - for undesirable performance variations
 - the **conformance** of the process with respect to certain rules or expected behaviors

- **Online Process Monitoring**

- Covers **currently running** process instances
- Produces **real-time** pictures of
 - the **performance** of ongoing cases
- Generate **alarms** or trigger **counteractions**
 - when certain **performance objectives** or **compliance rules** are **not fulfilled**

Granularity of Process Analysis



zur Mühlen (2004)

Process Monitoring Dimensions

Attribute	Possible Values				
Focus	Technical Information		Business Information		
Presentation	Active		Passive		
Timeframe	Running Processes		Completed Processes		
Aggregation	Single Instance		Multiple Instances		
Data Scope	Process	Process + Business Objects		Enterprise	
Object	Event	Activity	Process	Resource	Business Object
Process Scope	Activity	Segment	Process	Process Chain	

Presentation

Information

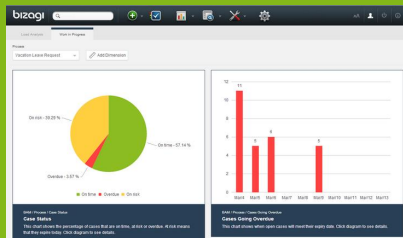
Statistics-based Process Monitoring with Dashboards



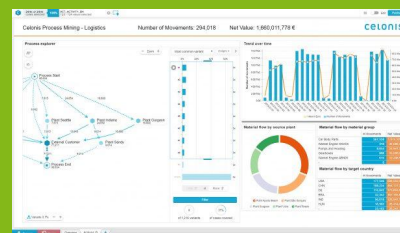
Types of Process Dashboards

- Process dashboards are often provided **off-the-shelf** by BPMS

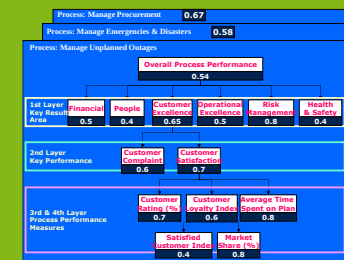
Operational
dashboards
(real-time)



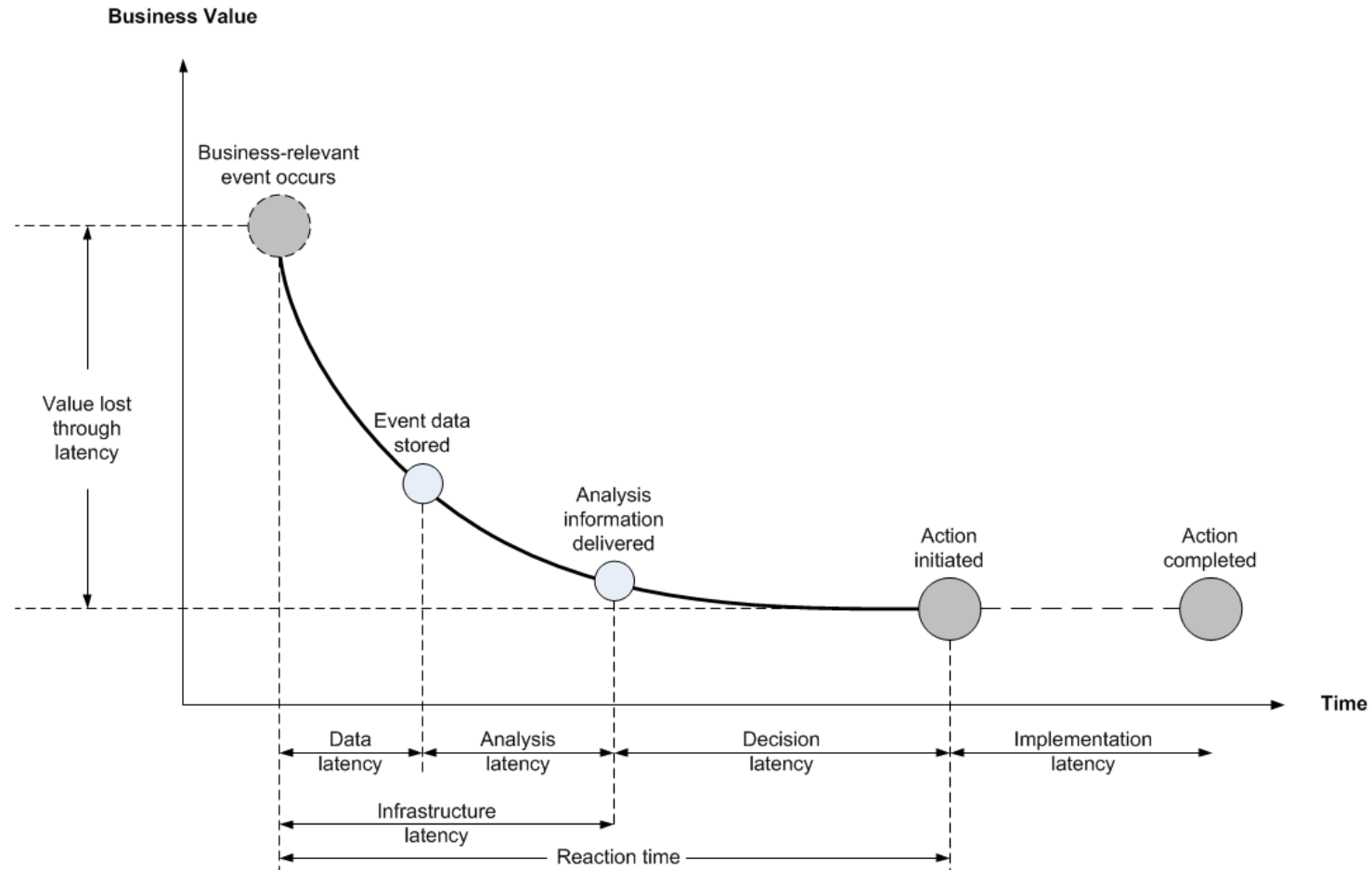
Tactical
dashboards
(historical)



Strategic
dashboards
(historical)



Information Latency



zur Mühlen, Shapiro (2010)

Operational Dashboards

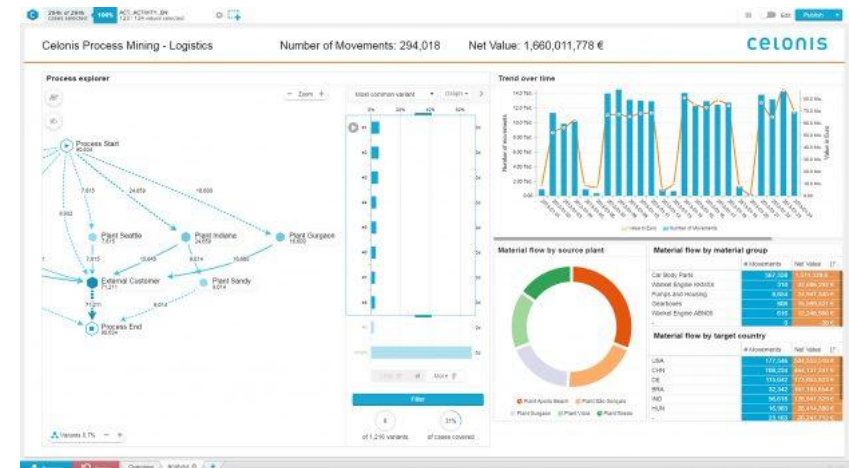
- Aimed at
 - Process **participants** and **operational managers**
 - **Ongoing** or recently completed cases
- Emphasis on **detect-and-respond**
 - E.g., work-in-progress, resource load, problematic cases (e.g., overdue/at-risk cases)
- Focus on
 - Level of resources
 - Tasks rather than cases



Tactical Dashboards

- Aimed at
 - Process **owners** and **functional managers, process analysts**
 - Process over a **relatively long period of time** (e.g., month, quarter, year)
- Emphasis on **analysis and management**
 - E.g., detecting bottlenecks
- Focus on
 - Process level (may be decomposable to task level)

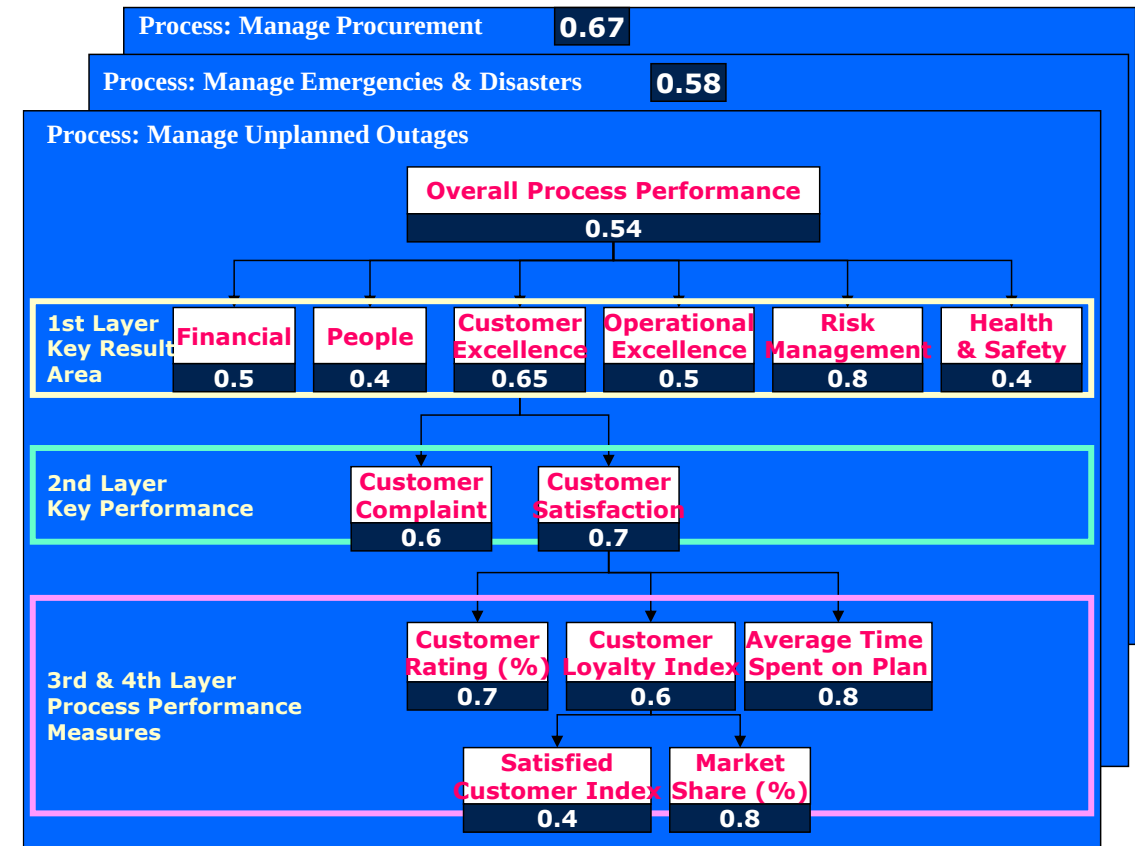
- **Longitudinal view**
 - Variation of a given performance measure over time
- **Cross-sectional view**
 - Distribution of a performance measure according to a given classification of cases



Dumas et al. (2018)

Strategic Dashboards

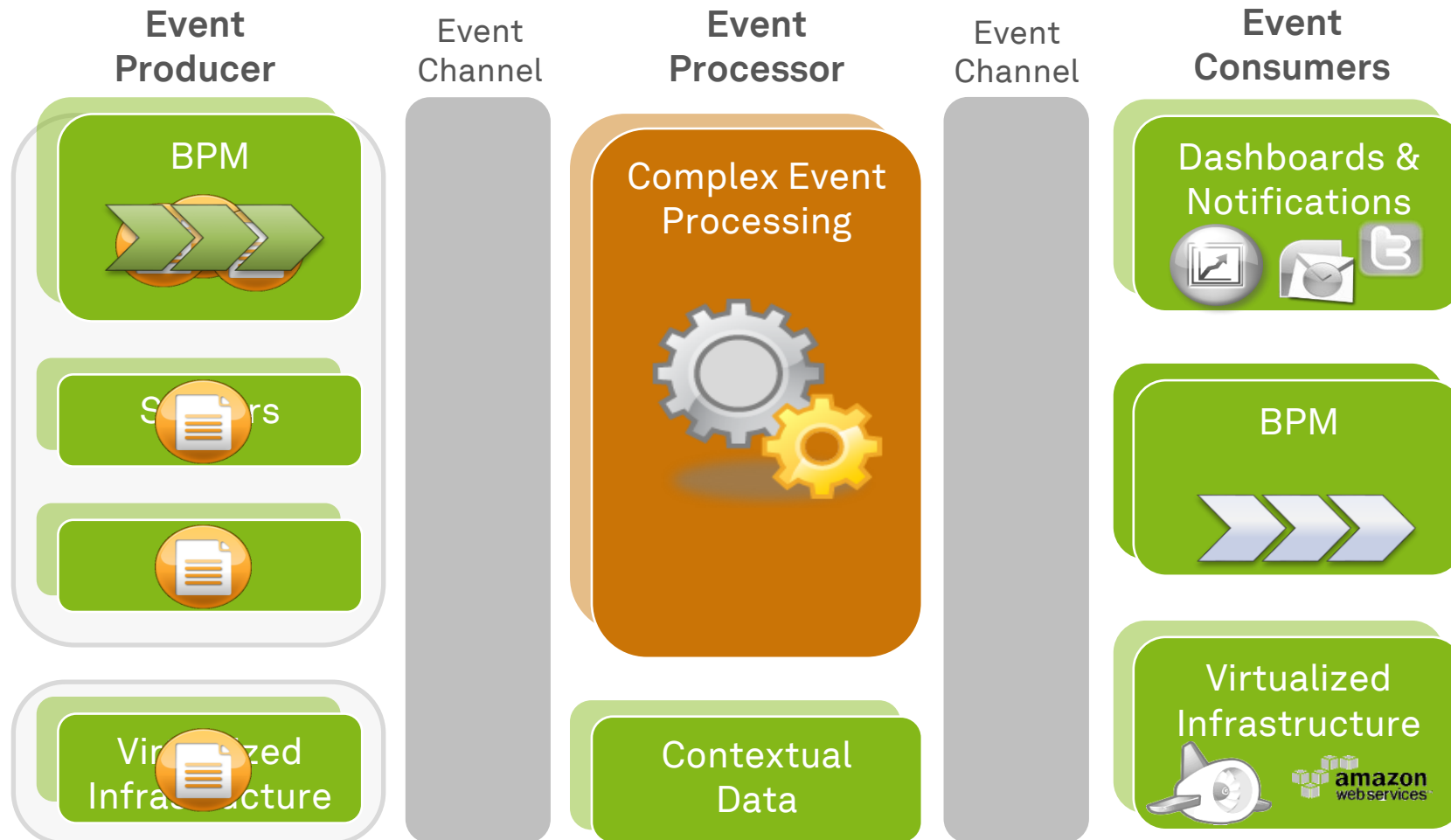
- Aimed at
 - **Executive managers**
- Emphasis on **high-level** performance and **link to strategic objectives**
 - Aggregating individual processes into entire **groups of processes**
 - Aggregating multiple performance measures into a **single measure**
- Focus on
 - Groups of processes
 - Performance dimensions



Monitoring Multiple BPMS with Business Activity Monitoring

- **BAM** is **Gartner's** term defining
 - how we can provide **real-time access** to critical business performance indicators
 - to improve the **speed and effectiveness of business operations**
- Unlike traditional real-time monitoring, BAM draws its information from
 - **multiple application systems** and
 - other **internal** and **external** (inter-enterprise) sources
 - **enabling** a broader and richer view of business activities
- **There is no universal BAM software with a process focus on the market**

Real-time Event-driven Architecture



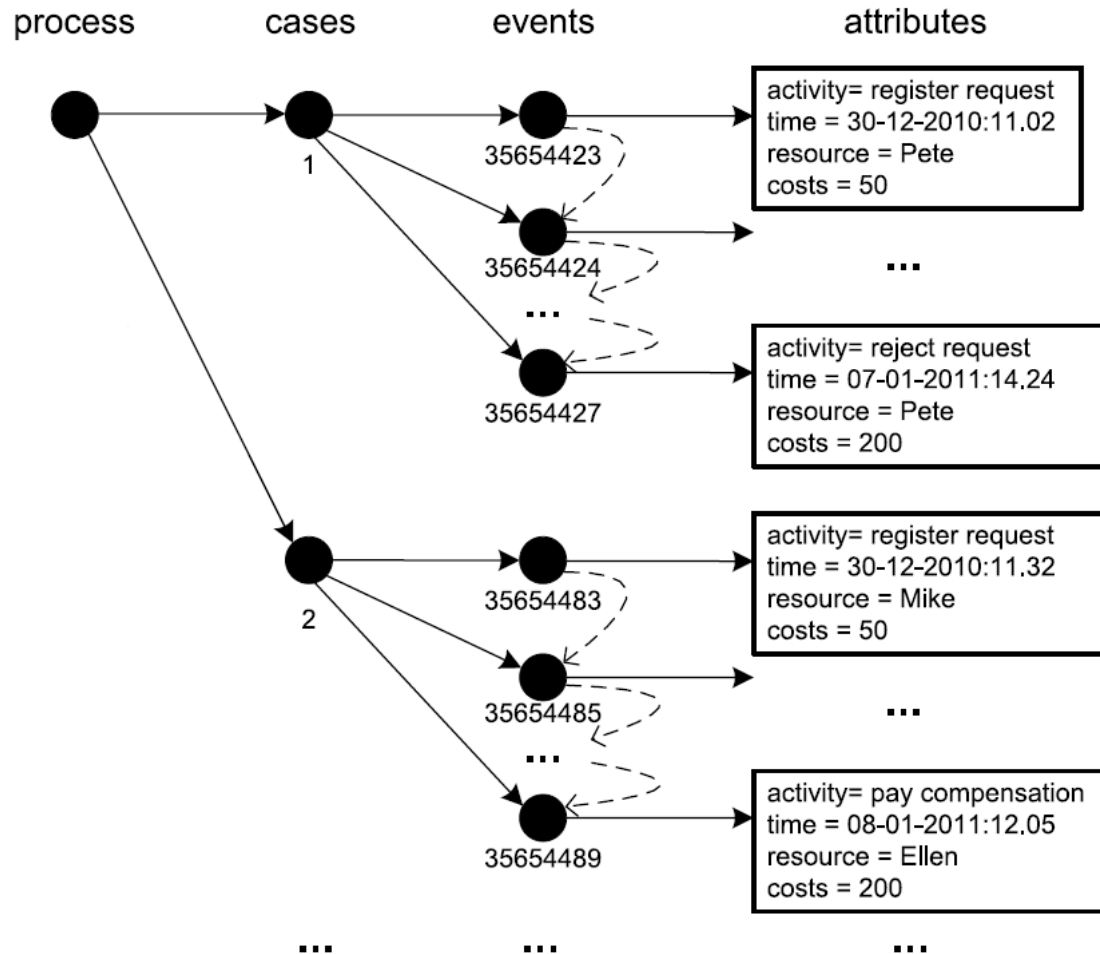
Model-based Process Monitoring with Process Mining



Process Mining

- “Process mining aims to **exploit event data** in a meaningful way”
- “Process mining techniques use event data to **discover** processes, check **compliance**, analyze **bottlenecks**, compare process **variants**, and suggest improvements”
- “Process mining adds the **process perspective** to machine learning and data mining”
- “Process mining brings together **traditional model-based process analysis** and **data-centric analysis techniques**”
- **Process mining can be performed exploratively or question-driven**

Structure of an Event Log



Event Record

Minimum attributes:
Case ID, timestamp, event class

Event Log

Case ID	Event ID	Timestamp	Activity	Resource
1	Ch-4680555556-1	2012-07-30 11:14	Check stock availability	SYS1
1	Re-5972222222-1	2012-07-30 14:20	Retrieve product	Chuck
1	Co-6319444444-1	2012-07-30 15:10	Confirm order	Chuck
1	Ge-6402777778-1	2012-07-30 15:22	Get shipping address	SYS2
1	Em-6555555556-1	2012-07-30 15:44	Emit invoice	SYS2
1	Re-4180555556-1	2012-08-04 10:02	Receive payment	SYS2
1	Sh-4659722222-1	2012-08-05 11:11	Ship product	Susi
1	Ar-3833333333-1	2012-08-06 09:12	Archive order	DMS
2	Ch-4055555556-2	2012-08-01 09:44	Check stock availability	SYS1
2	Ch-4208333333-2	2012-08-01 10:06	Check materials availability	SYS1
2	Re-4666666667-2	2012-08-01 11:12	Request raw materials	Ringo
2	Ob-3263888889-2	2012-08-03 07:50	Obtain raw materials	Olaf
2	Ma-6131944444-2	2012-08-04 14:43	Manufacture product	SYS1
2	Co-6187615741-2	2012-08-04 14:51	Confirm order	Conny
2	Em-6388888889-2	2012-08-04 15:20	Emit invoice	SYS2
2	Ge-6439814815-2	2012-08-04 15:27	Get shipping address	SYS2
2	Sh-7277777778-2	2012-08-04 17:28	Ship product	Sara
2	Re-3611111111-2	2012-08-07 08:40	Receive payment	SYS2
2	Ar-3680555556-2	2012-08-07 08:50	Archive order	DMS
3	Ch-4208333333-3	2012-08-02 10:06	Check stock availability	SYS1
3	Ch-4243055556-3	2012-08-02 10:11	Check materials availability	SYS1
3	Ma-6694444444-3	2012-08-02 16:04	Manufacture product	SYS1
3	Co-6751157407-3	2012-08-02 16:12	Confirm order	Chuck

Trace
(ordered
sequence)

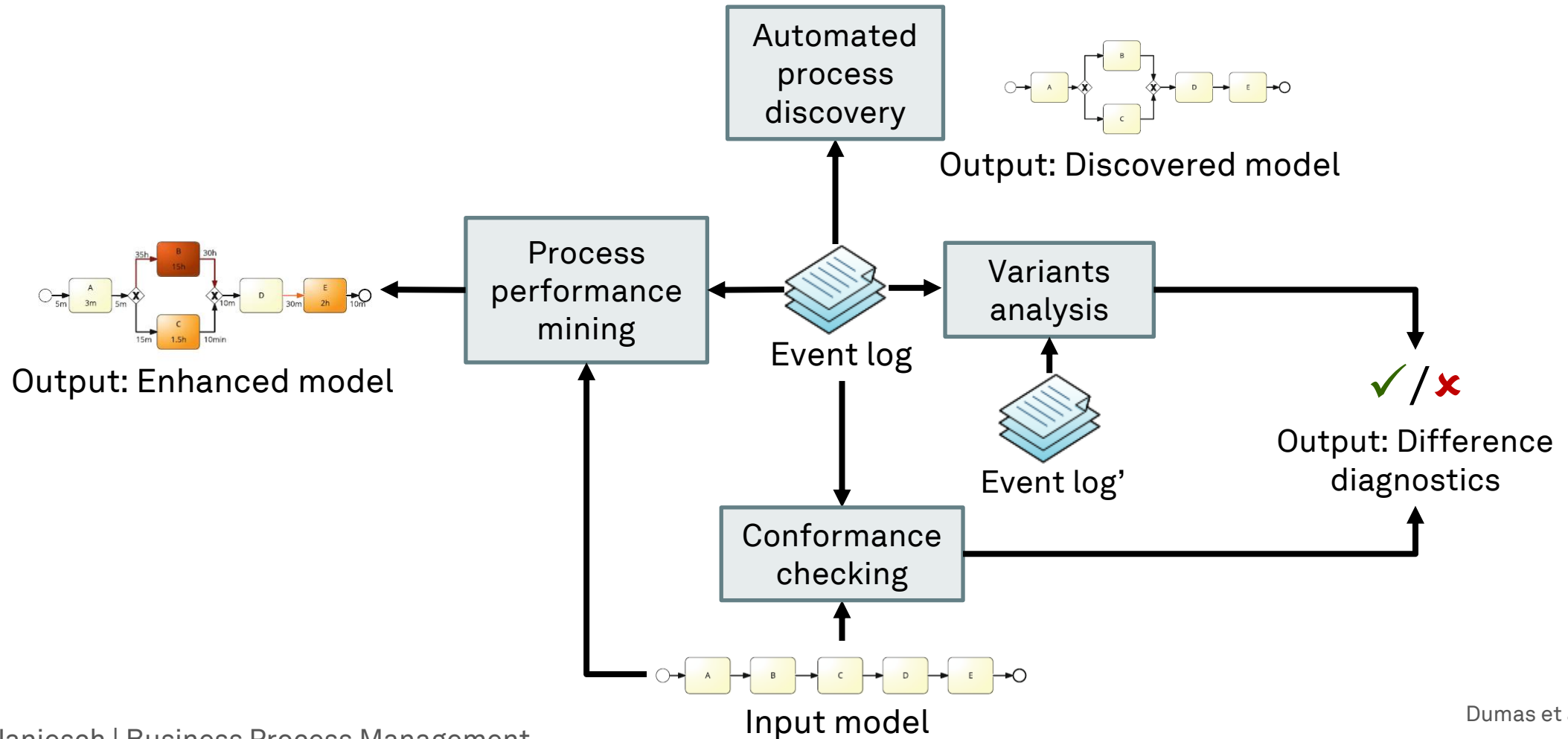
van der Aalst (2016), Dumas et al. (2018)

Process Mining Challenges

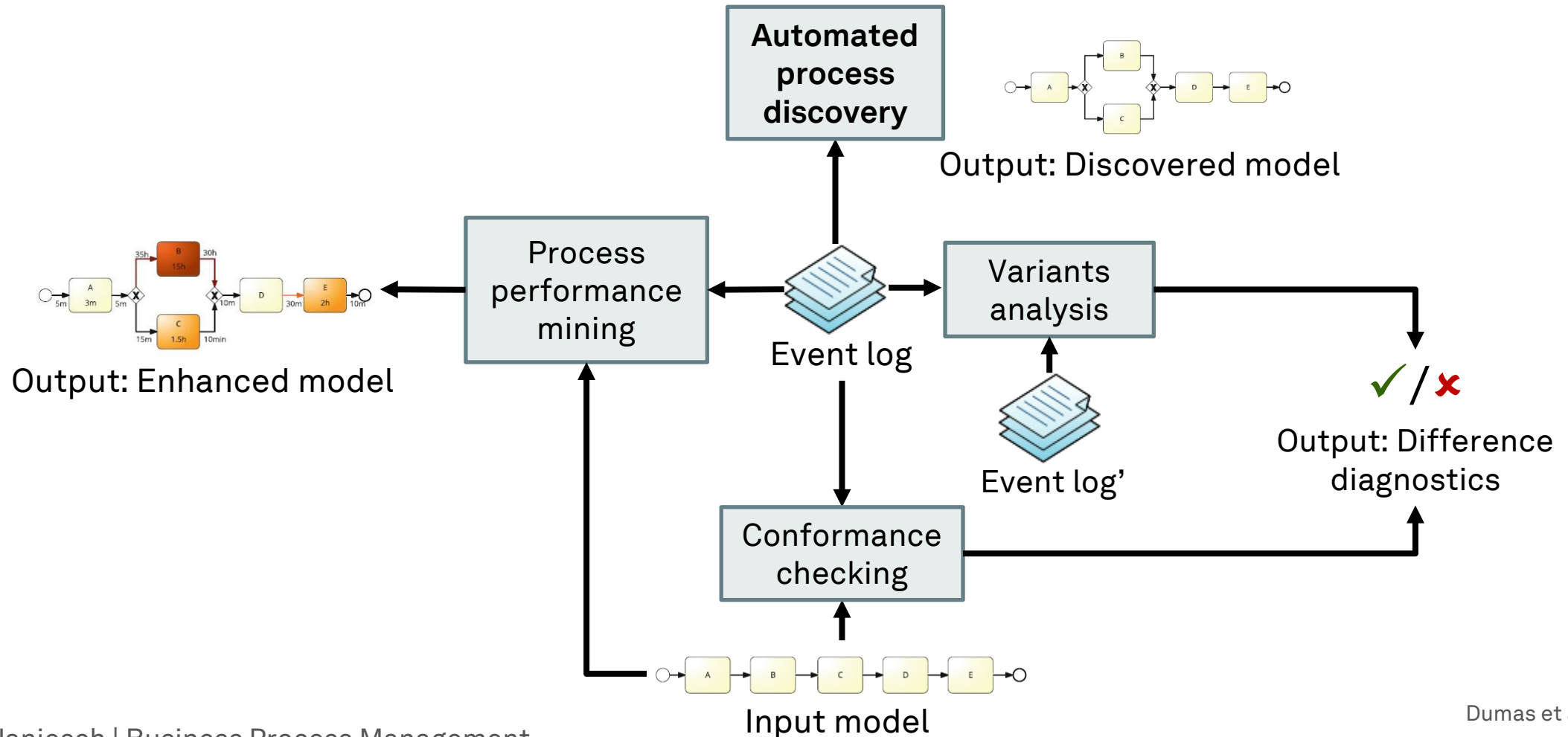
- **Correlation** challenge
 - Identify the case an event belongs to
- **Timestamps** challenge
 - Logging is often delayed until the system has idle time: sequential events with the same timestamp, logs from different BPMS/time zones
- **Longevity** challenge
 - Long running processes might be too slow for snapshot window
- **Granularity** challenge
 - Abstraction of model might be different from log
- **Scoping** challenge
 - Information system does not directly produce event logs, logs are synthesized

Dumas et al. (2013, 2018)

Process Mining Techniques



Process Mining Techniques



Automated Process Discovery

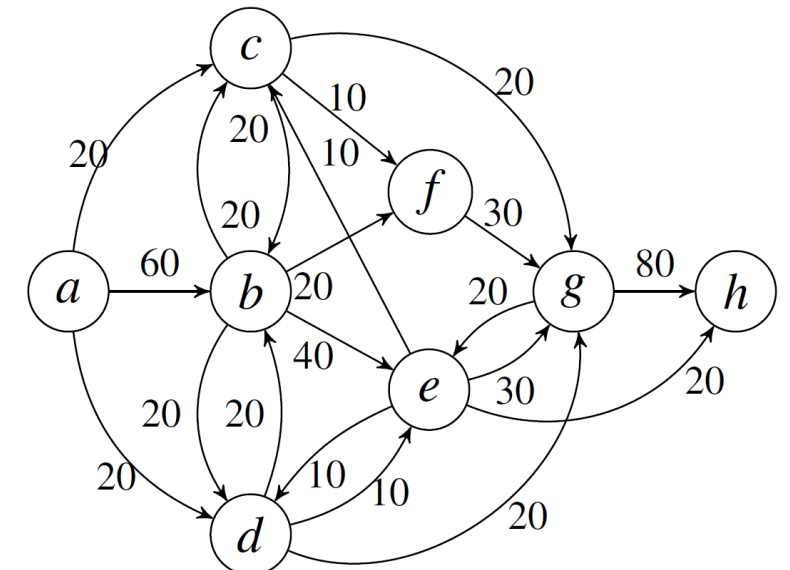
- Goal
 - Construct a process model that **captures the behavior** of an event log in a **representative way**
- Construction
 - **Automatically and generically** using an algorithm
 - Should make minimal assumptions about properties of the log and the resulting process model
- Result
 - Constructed process model should be able to **replay** the cases of the event log and
 - **Forbid behavior** not found in the logs

Dependency Graph

- aka **directly-follows graph** aka **process map**
- Each **activity** is represented by one node
- An **arc** from activity A to activity B means that B is directly followed by A in at least one trace in the log
- Arcs in a process map can be annotated with
 - Absolute/ relative frequency (percentage)
 - Temporal delay
- **Does not allow us to distinguish whether two tasks are in parallel, in a loop, or in some other relation**

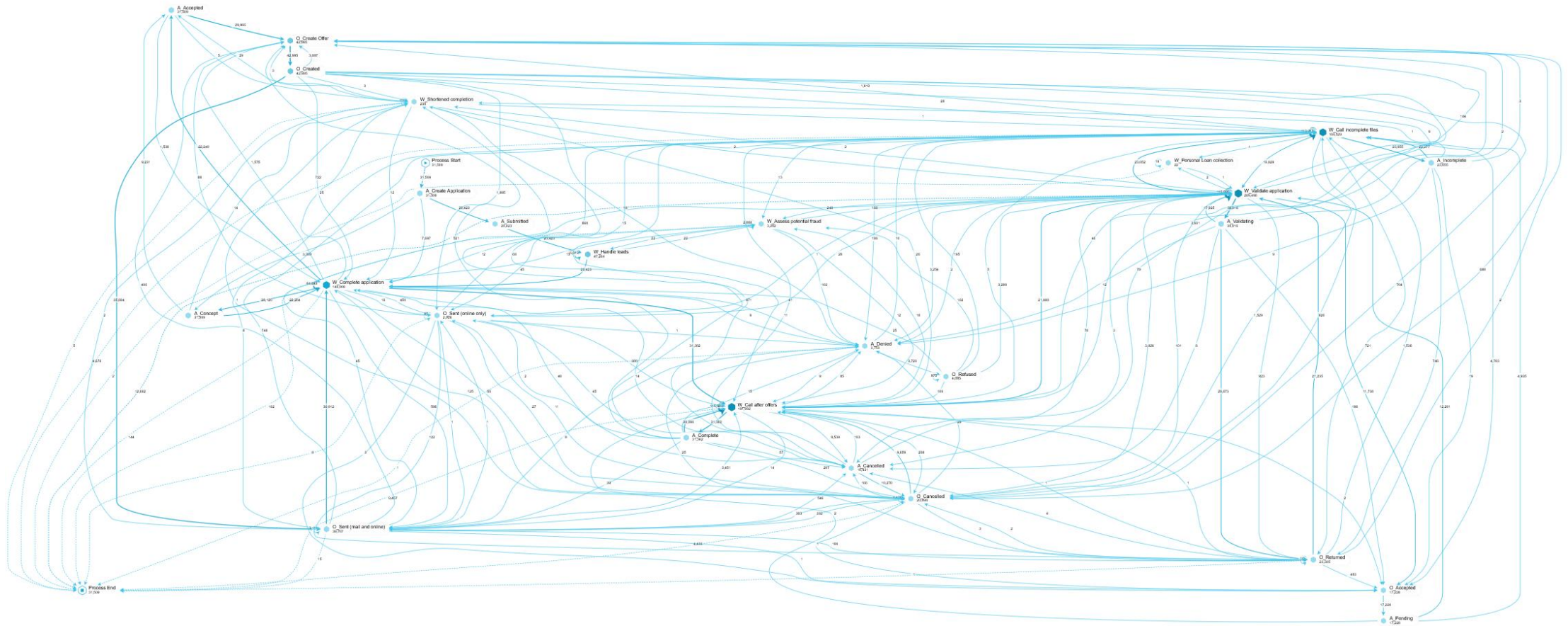
Event log:

10: a,b,c,g,e,h	10: a,b,e,d,g,h
10: a,b,c,f,g,h	10: a,c,b,e,g,h
10: a,b,d,g,e,h	10: a,c,b,f,g,h
10: a,b,d,e,g,h	10: a,d,b,e,g,h
10: a,b,e,c,g,h	10: a,d,b,f,g,h



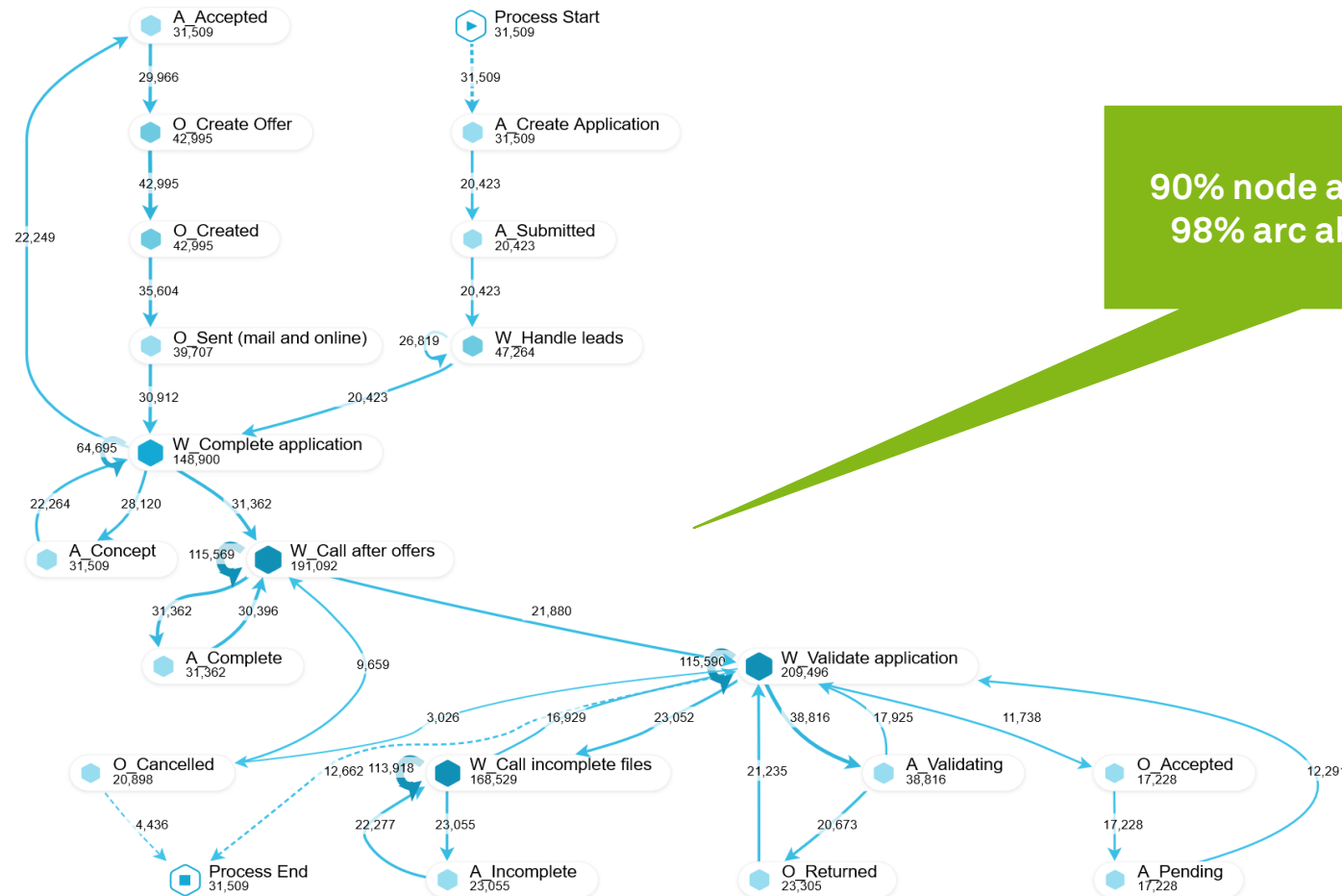
Dumas et al. (2018)

Real-life Process Map in Celonis



Dumas et al. (2018)

Real-life Process Map in Celonis (Filtered)



90% node abstraction,
98% arc abstraction

Abstraction and Filtering

- Abstraction
 - **Only show** the most frequent **nodes** or **arcs** of a dependency graph
- Event Log Filtering
 - Only show traces that **fulfil** (or do not fulfil) **a condition**
 - **Event filters**
 - Retain only **events** that fulfil a condition (e.g., certain event class)
 - **Event pair filters**
 - Retain only **events** that fulfil a condition **in respect to other events** (e.g., follows)
 - **Trace filters**
 - Retain only **traces** that fulfil a given condition (e.g., contain certain events or endpoints, in relation to performance measures)

Overview of Discovery Algorithms

- **Alpha miner** (α -miner)
 - Simple, limited, not robust
- **Heuristics miner**
 - Uses frequencies, detects self- & short-loops, handles incomplete and noisy logs
 - **Structured heuristic miner** further converts models to block-structured models
- **Inductive miner**
 - Divides dependency graphs in block structures using a different approach
- **Split miner**
 - Discovers sound but not always block-structured models
- **Genetic miner**
 - Iterative, non-deterministic procedure

van der Aalst (2016), Dumas et al. (2018)

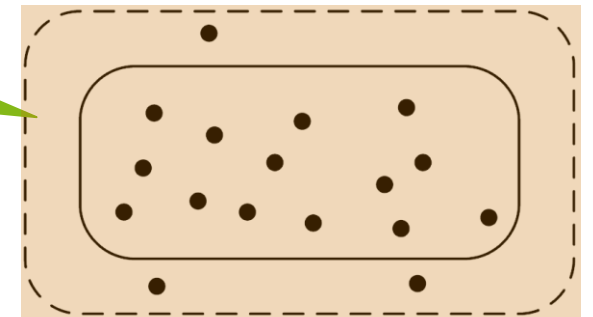
Characteristics of Advanced Process Discovery Algorithms

- **Representational Bias**
 - Inability to represent **concurrency**
 - Inability to deal with (arbitrary) **loops**
 - Inability to represent **silent actions**
 - Inability to represent **duplicate actions**
 - Inability to model (inclusive) **OR-splits/ joins**
 - Inability to represent **non-free-choice behavior**
 - Inability to represent **hierarchy**
- **Ability to Deal With Noise**
 - Deal with **errors** and **rare** behavior
- **Completeness Notion Assumed**
 - Too weak → underfitting
 - Too strong (e.g., log contains all possible traces) → overfitting
- **Approach Used**

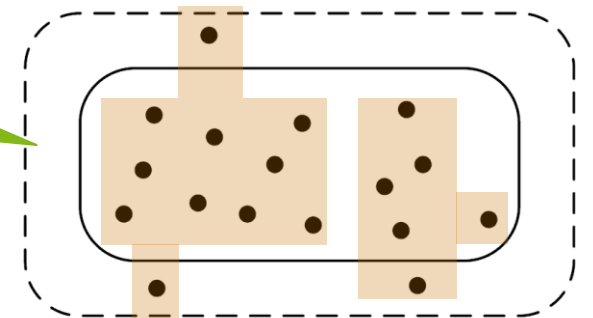
Quality Measures for Automated Process Discovery (1)

- **Fitness**
 - Ability to replay behavior found in the logs
- **Precision**
 - Degree of behavior allowed by the model, but not in the logs
 - Do not **underfit**
- **Generalization**
 - Work with incomplete behavior
 - Do not **overfit**
- **Simplicity/ Complexity (Occam's razor)**
 - Difficulty of understanding the model (e.g., number of nodes, arcs, etc.)
 - **The simplest model to explain the log's behavior, is the best model**

Underfitting model

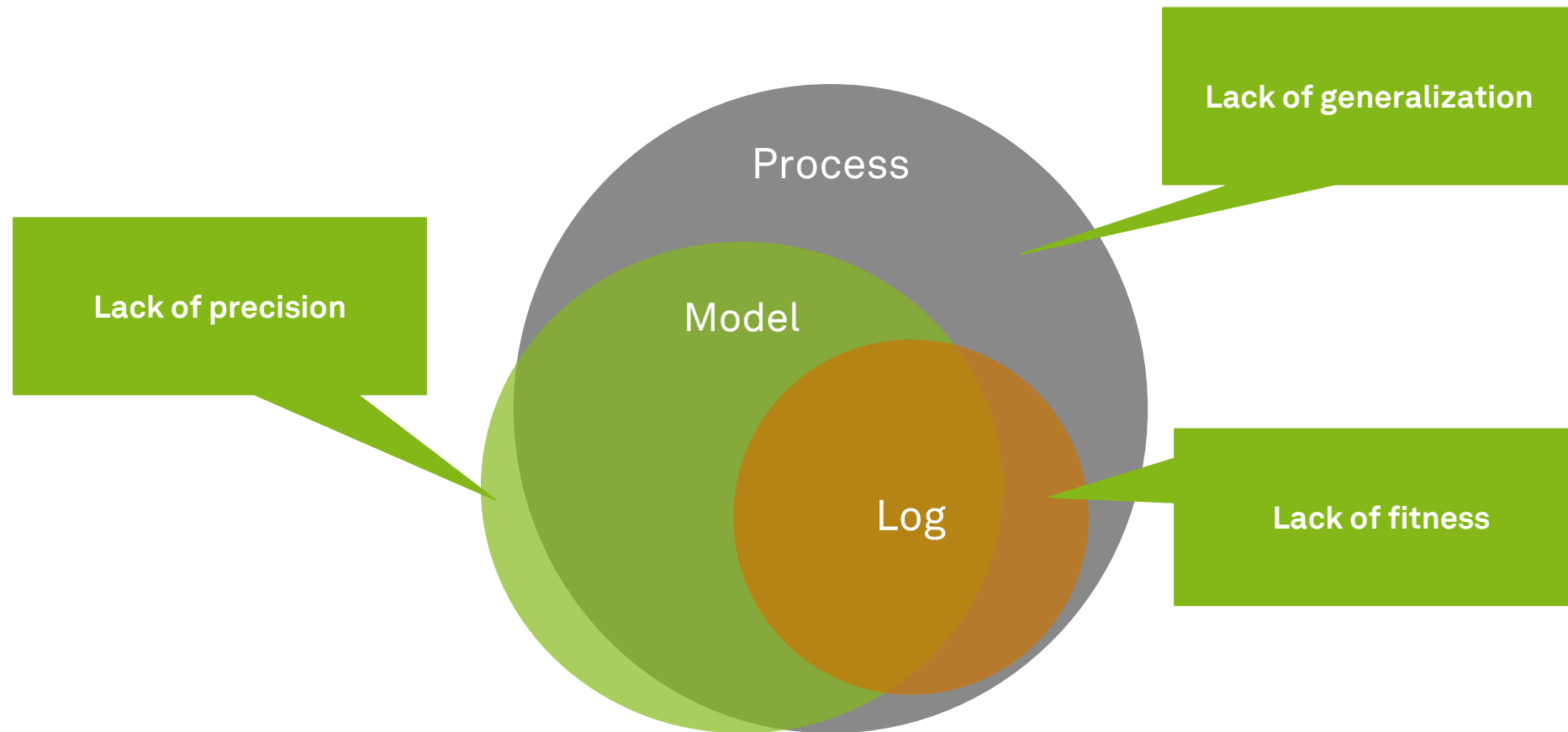


Overfitting model

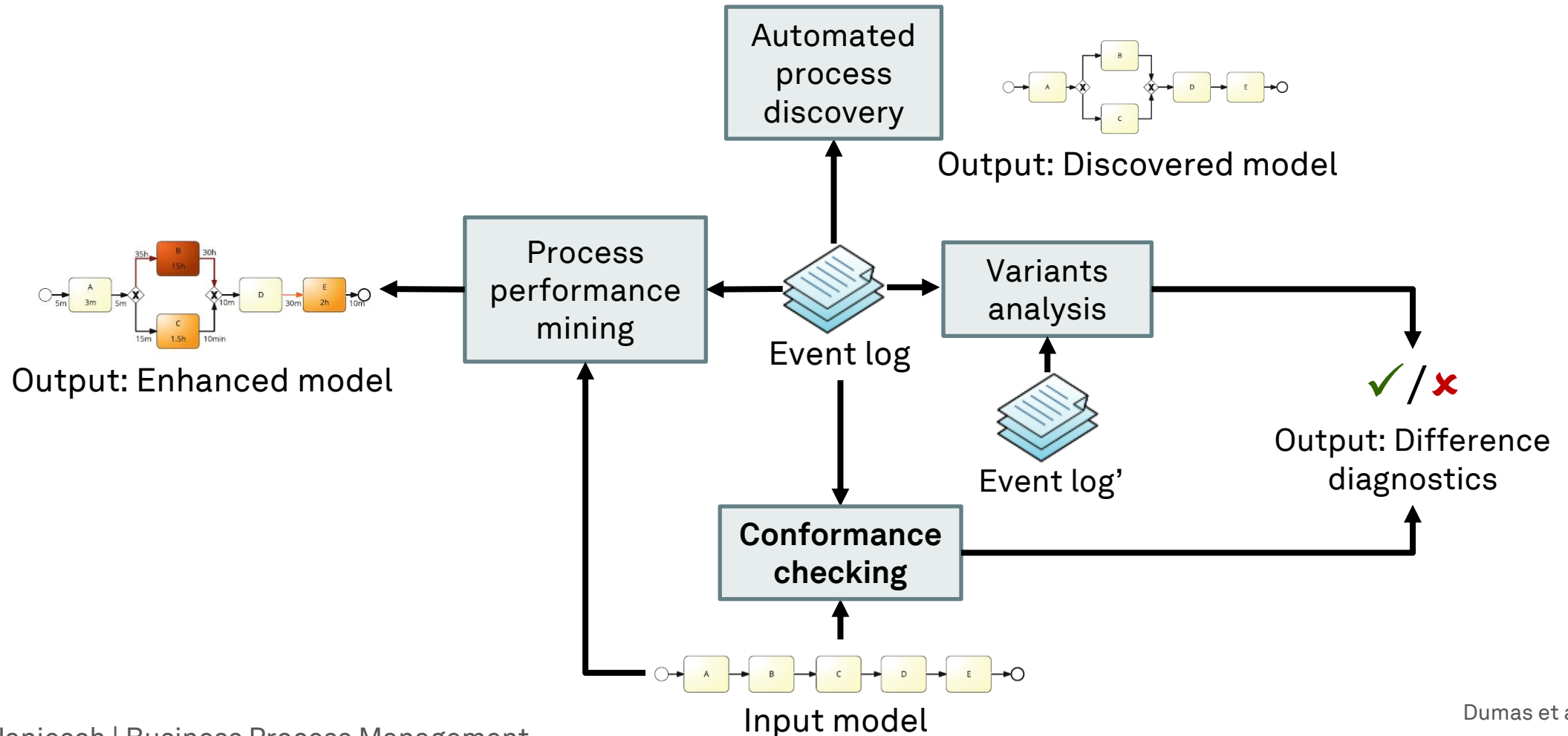


van der Aalst, (2016), Dumas et al. (2013, 2018)

Quality Measures for Automated Process Discovery (2)



Process Mining Techniques



Conformance Checking

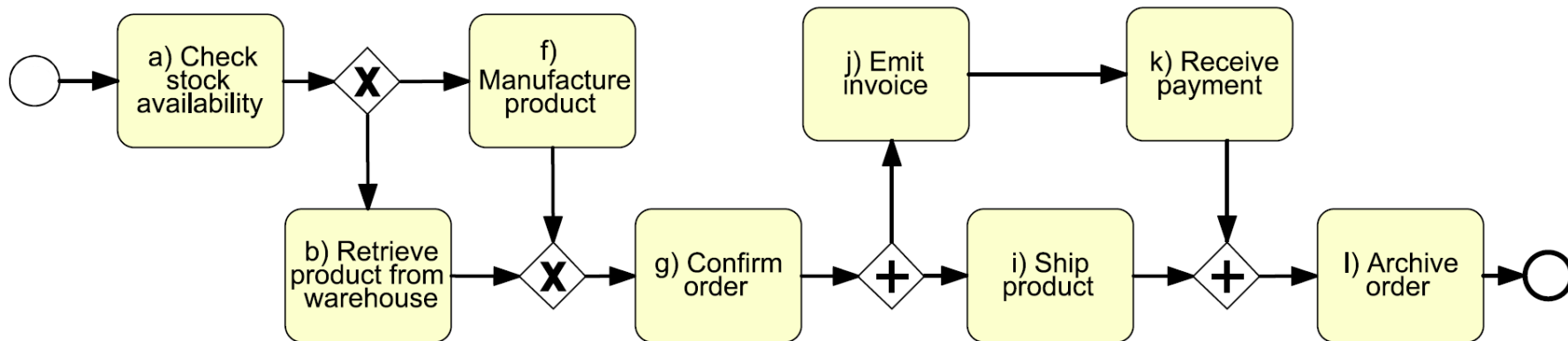
- Goal
 - Determine whether the execution of a process **follows predefined rules or constraints**, or if it **violates** them
- Analysis
 - With families of conformance checking **techniques**
 - Replay, trace alignment, and behavioral alignment
 - Explicit **constraints** or normative process **model**
 - Perspectives: control flow, data, and resources
- Result
 - **Violations** that relate to one of the three process perspectives in isolation or in combination

Conformance Checking of Control Flow

- **Explicit Constraints**
 - **Mandatoriness constraint**
 - Tasks are required from a control perspective
 - e.g., “approval task”
 - **Exclusiveness constraint**
 - Tasks that relate to a decision
 - Two or more tasks cannot co-occur in the same case
 - e.g., “buy” vs. “rent”
 - **Ordering constraint**
 - Two tasks must always occur in a specific order
 - E.g., “approval” → “order”
- **Normative process model**
 - **Unfitting** log behavior
 - Behavior observed in the log that is not allowed by the model
 - **Additional** model behavior
 - Behavior allowed in the model but never observed in the log

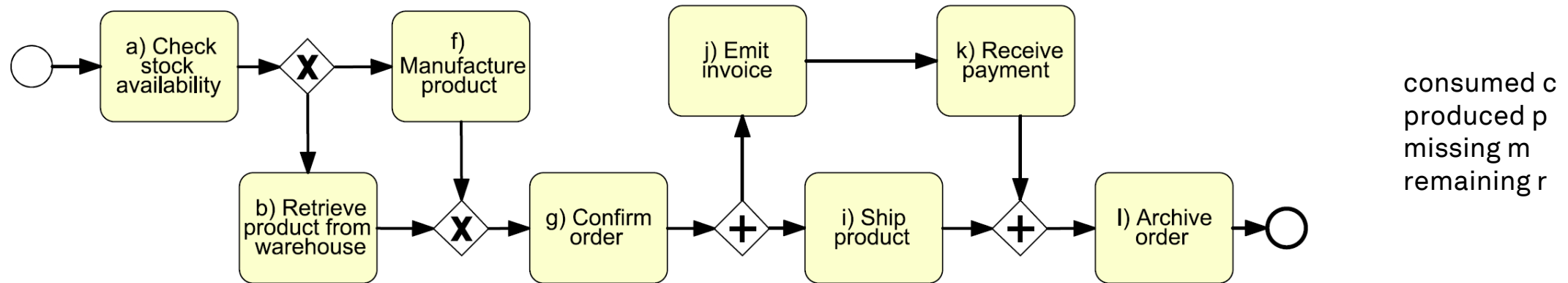
Normative Process Model Replay

- For each trace in the workflow log
 - Replay with tokens
 - Record at each step whether an activity was **allowed** to be executed according to the rules of the model
- Is (a, b, g, j, i, k, l) valid?



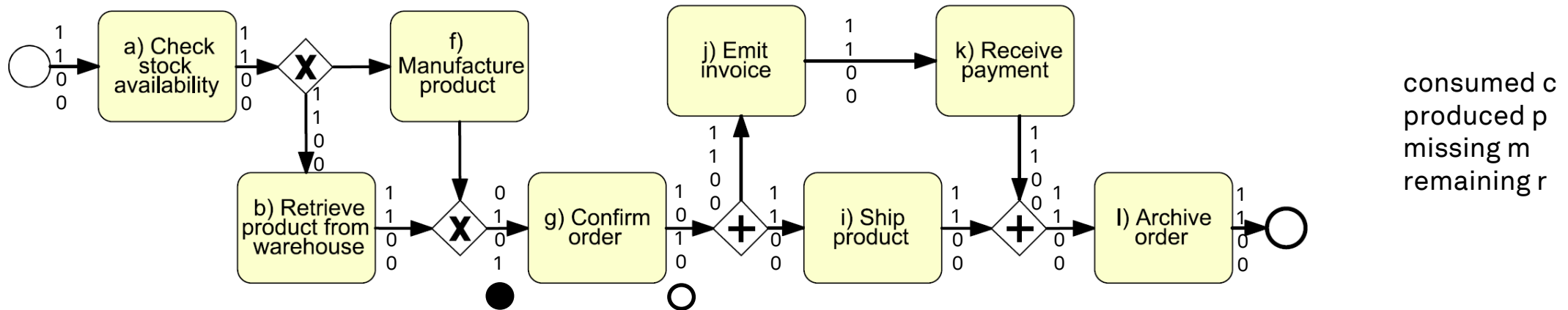
- Is (a, b, i, j, k, l) valid, too? If not, what does it violate?

Replay of Tokens in a Normative Process Model



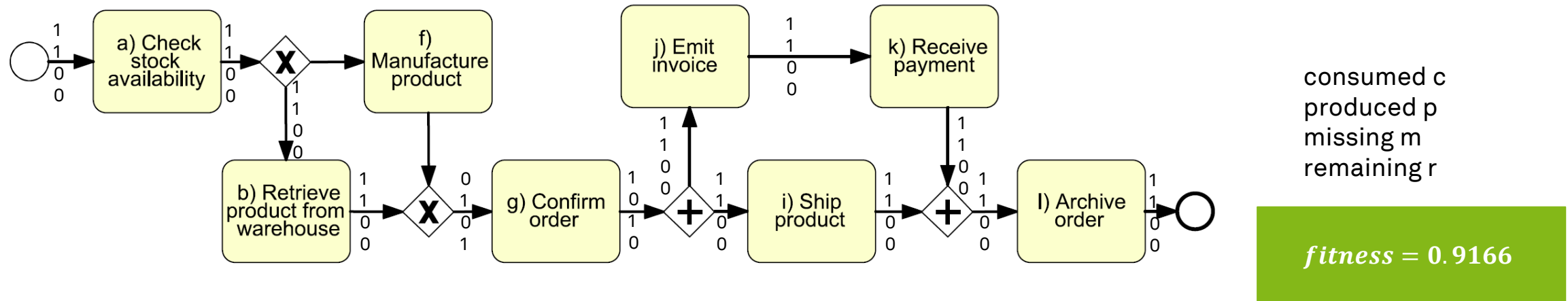
- How many misplaced tokens in (a, b, i, j, k, l)?
 - Number of tokens that are **correctly consumed c**
 - Number of tokens that are **correctly produced p**
 - Number of tokens that are **missing for executing m** the next activity in the log
 - Number of tokens **remaining unconsumed r** after executing the final activity in the log

Replay of Tokens in a Normative Process Model



- How many misplaced tokens in (a, b, i, j, k, l)?
 - Number of tokens that are **correctly consumed c**
 - Number of tokens that are **correctly produced p**
 - Number of tokens that are **missing for executing m** the next activity in the log
 - Number of tokens **remaining unconsumed r** after executing the final activity in the log

Replay of Tokens in a Normative Process Model



- How many misplaced tokens in (a, b, i, j, k, l)?
 - Number of tokens that are **correctly consumed c (12)**
 - Number of tokens that are **correctly produced p (12)**
 - Number of tokens that are **missing for executing m** the next activity in the log **(1)**
 - Number of tokens **remaining unconsumed r** after executing the final activity in the log **(1)**

$$fitness = \frac{1}{2} \left(1 - \frac{m}{c} \right) + \frac{1}{2} \left(1 - \frac{r}{p} \right)$$

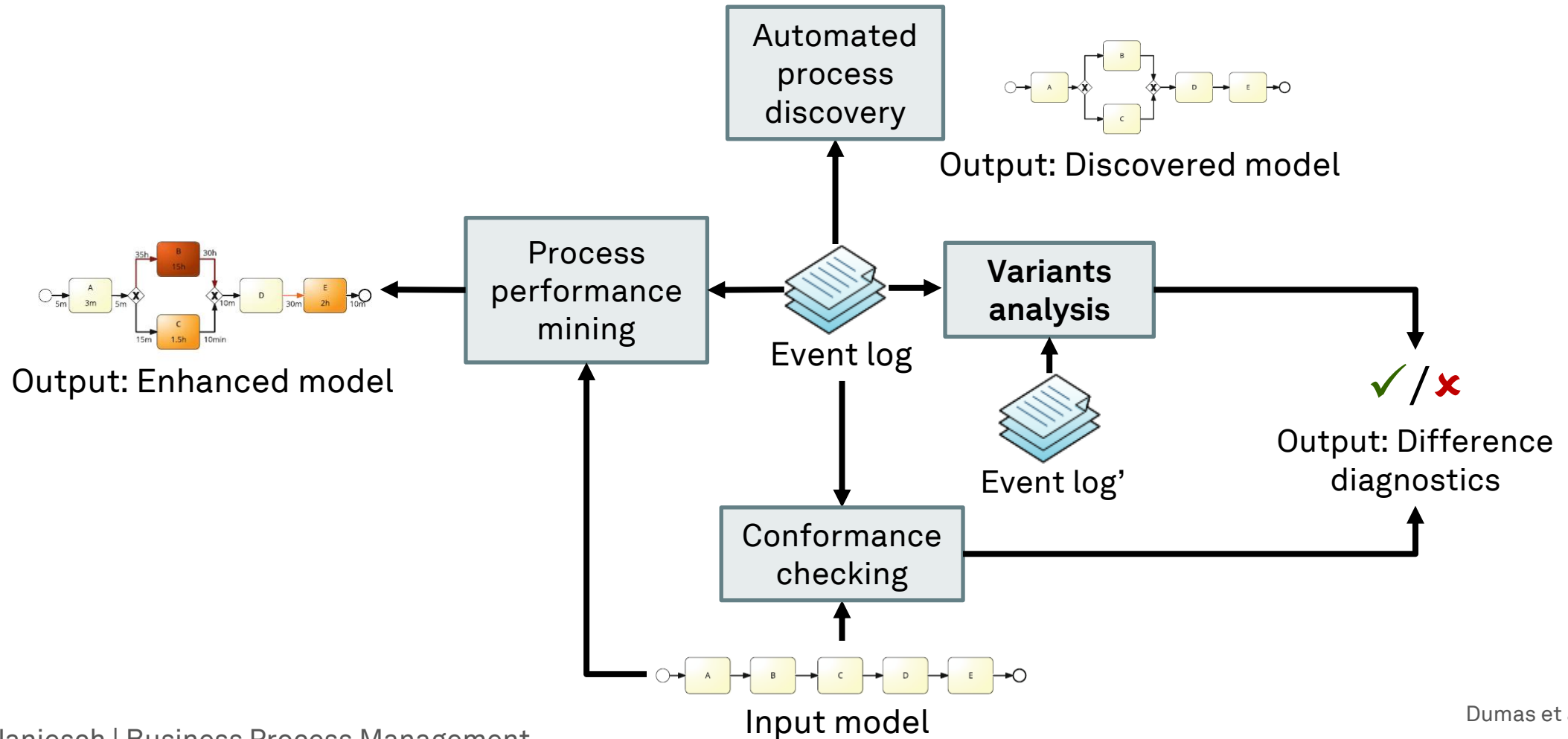
Trace Alignment and Behavioral Alignment

- With **replay**, the recovery from an error is done locally
- Replay does **not identify all errors at once** that explain the unfitting trace
- **Trace Alignment**
 - Identifies the **closest corresponding trace** that can be parsed by the model
 - Shows the points of divergence
 - **Does not explicitly handle concurrent tasks nor cyclic behavior**
- **Behavioral Alignment**
 - Aligns the entire event log against the process model at once
 - Identifies differences that cannot be observed at individual traces level

Conformance of Data and Resources

- Participants usually require **permissions** to execute certain activities
- Permissions are bundled for specific **roles**
- Violations of permissions can be checked by searching for each **activity conducted** by a participant whether or not an **appropriate role or permission** existed
- Specific control rules which require two different persons to approve a business transaction are called **separation of duties** constraints
 - These rules do not necessarily involve supervisors

Process Mining Techniques

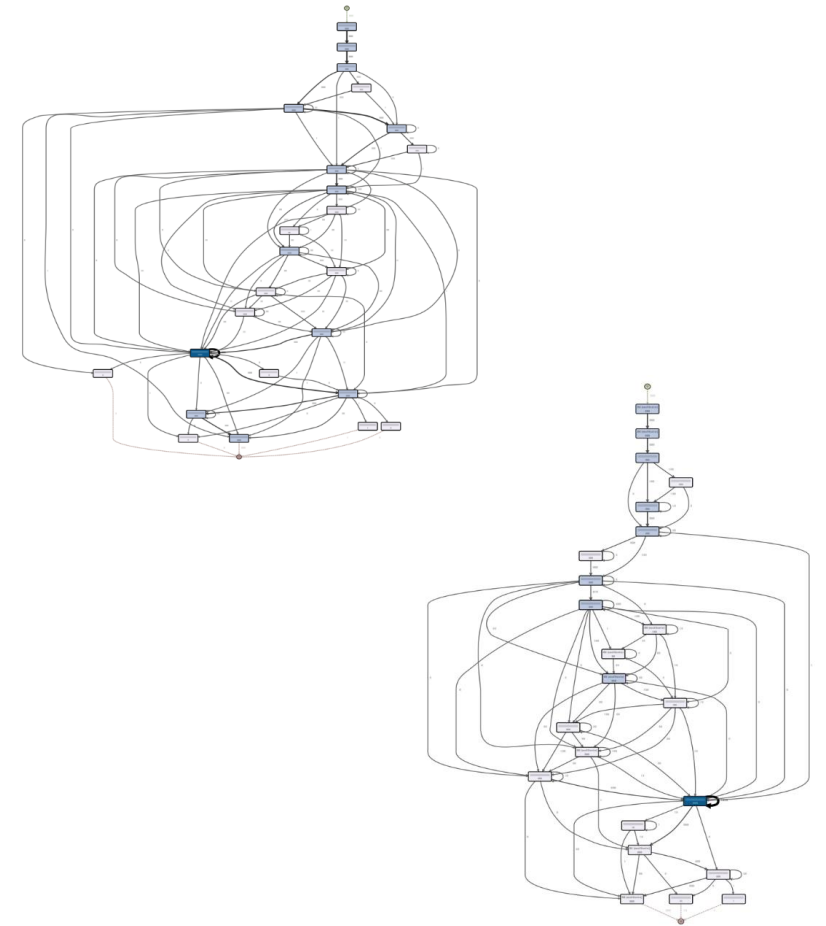


Variants Analysis

- Goal
 - Determine the **root cause for (performance) variation** between cases
- Analysis
 - Differences across **subsets of cases** in a process
 - Split based on time, resources, business data/attributes
 - Deviance mining: Comparison of normal performance with undesirable performance
- Result
 - **Characteristics** that are commonly found in the distinct cases

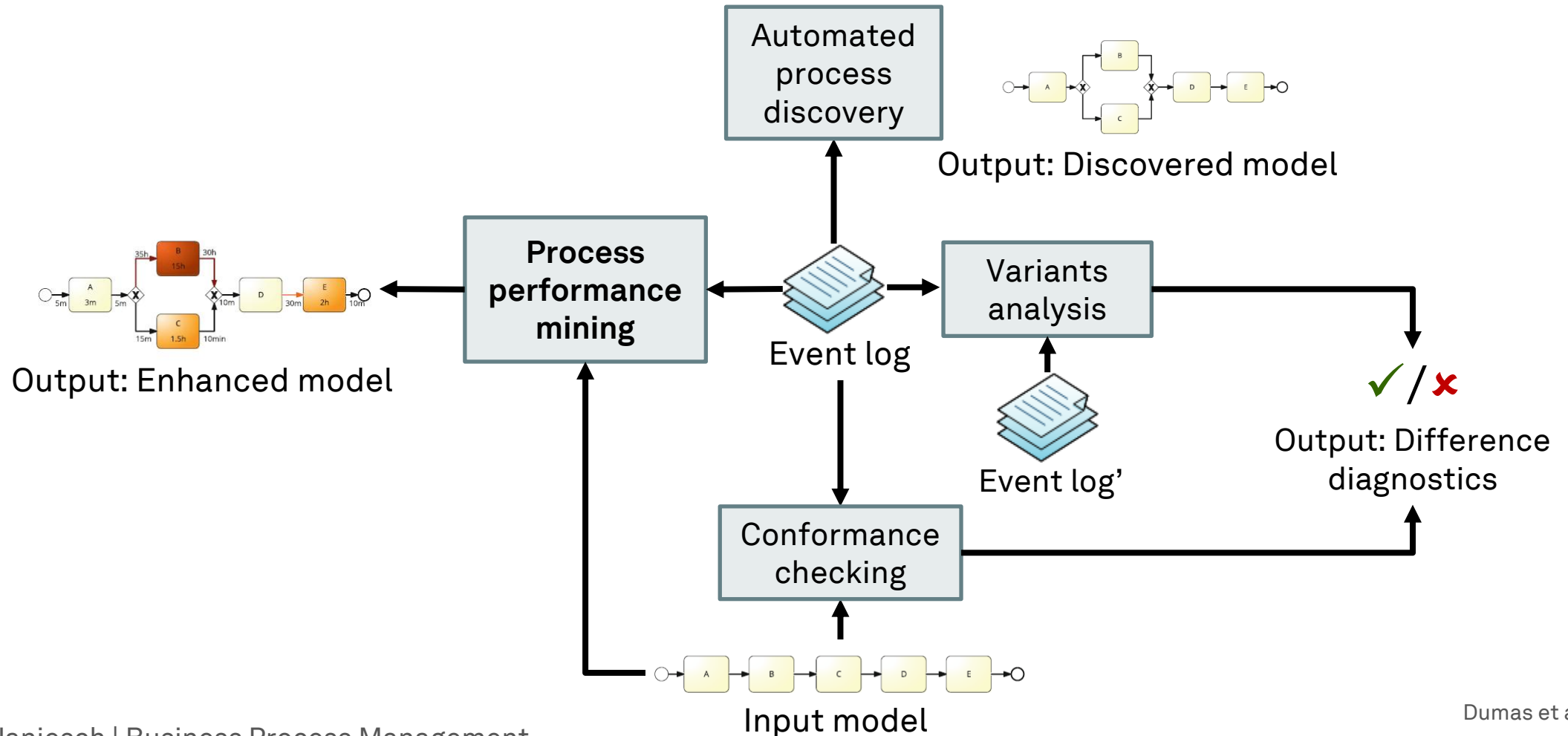
Variants Analysis

- **Manual Comparison**
 - Only suitable for small or filtered event logs
- **Log Delta Analysis**
 - Compares two event logs and produces a **set of patterns or statements** that are common in one event log and uncommon in the other event log
- **Association Rules**
 - Generally used to transform an event log into patterns
 - **Discriminative sequence mining** compares two event logs and identifies **association rules**, which are common in one event log and uncommon in the event other log



van Beest et al. (2015), Dumas et al. (2018)

Process Mining Techniques



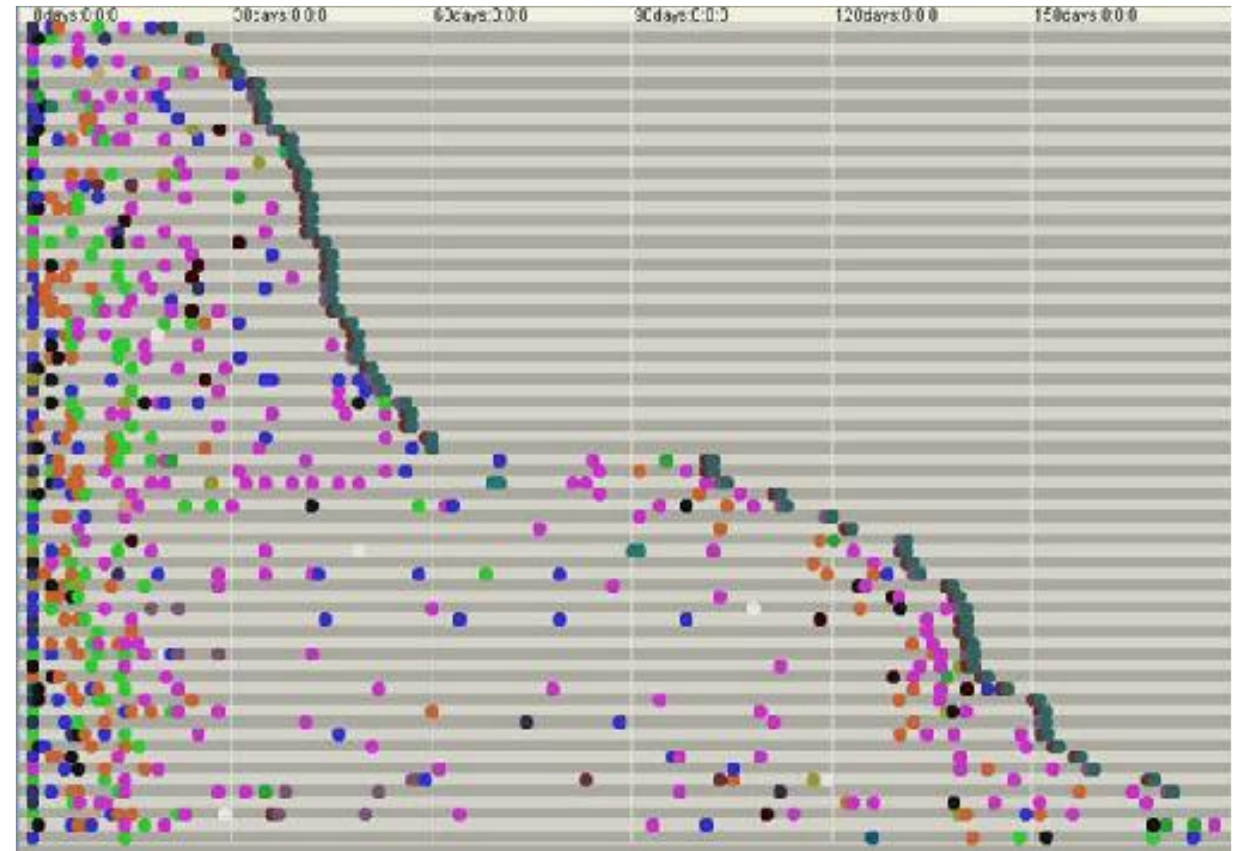
Process Performance Mining

- Goal
 - **Quantify and visualize** the performance of a process automatically
- Analysis
 - Can be organized around the devil's quadrangle of time, cost, quality, and flexibility
- Result
 - Quantified measures that evaluate a process

Process Performance Mining: Time

- **Dotted chart**

- One line per trace, each line contains points, one point per event
- Each event type is mapped to a color
- Position of the point denotes its occurrence time in (relative) scale
- Birds-eye view of the timing of different events (e.g., activity end times)
- **Does not allow one to see the *processing times of activities***

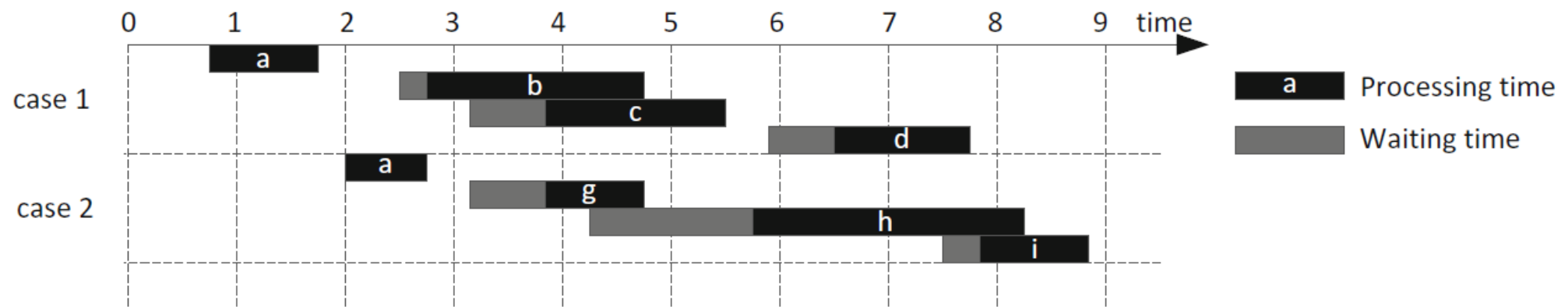


Dumas et al. (2018)

Process Performance Mining: Time

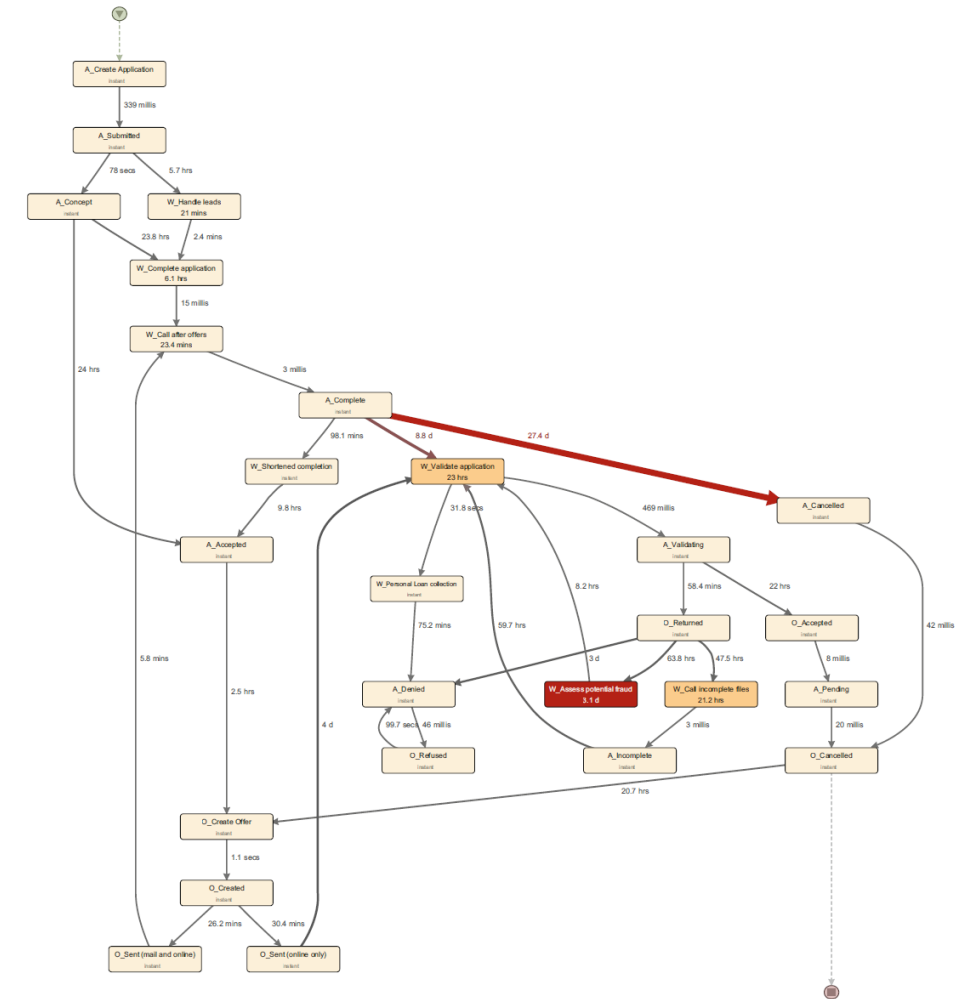
- **Timeline chart**

- One line per trace, each line contains segments capturing the start and end of tasks
- Captures process time (unlike dotted charts)
- Good to show “representative” traces
- **Not scalable for large event logs**



Process Performance Mining: Time

- **Performance-enhanced dependency graphs**
 - Nodes are color-coded with respect to a performance measure
 - Typically, edge color denotes waiting time, node color denotes processing time
 - Typically, nodes represent activities, but they may represent resources and then arcs denote hands-offs between resources



Process Performance Mining: Cost

- **Direct cost** of labor
 - Salaries, only 10% today in automated production
- **Overhead** cost of labor
 - Education & training, installation & maintenance, marketing & support
- **Activity-based Costing** allocates indirect cost to an activity's direct cost
 - All business activities must be broken down into their discrete components
 - Different cost (product, service, resource) can be determined based on **cost drivers**
 - Costs are assigned to adequate **activity centers** to determine total activity cost
 - Cost of activities are related to the products or processes to be measured

Process Performance Mining: Quality

- Quality is often **not directly visible** from event logs
- Repetitions are a good indication for a lack of quality
- They typically occur when a task has not been completed successfully

Cycle time of a
rework block

$$CT = \frac{T}{1 - r}$$

$$r = 1 - \frac{T}{CT}$$

Rework probability

- Alternative: track repetitions based on the assignment of tasks to resources (e.g., via ticketing system)

Process Performance Mining: Flexibility

- Compare the desired level of flexibility with the actual flexibility
- A **lack of flexibility** hints at a **lack of resilience**
- A **surplus of flexibility** hints at a **lack of standardization**
- Suggested Measures
 - **Distinct executions**
 - Traces DE with unique sequences in event log L
 - Issues with **concurrency**
 - **Degree of optionality**
 - Tasks T_{opt} in a sequence of tasks T are optional

$$DE = |L|$$

$$OPT = \frac{T_{opt}}{T}$$

Learning Goals of Today

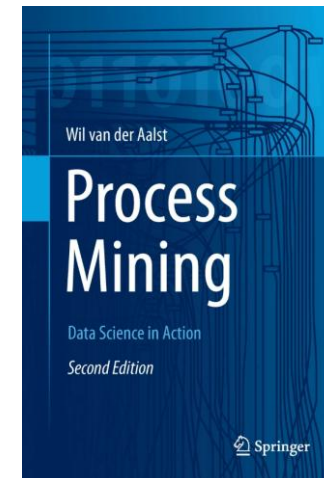
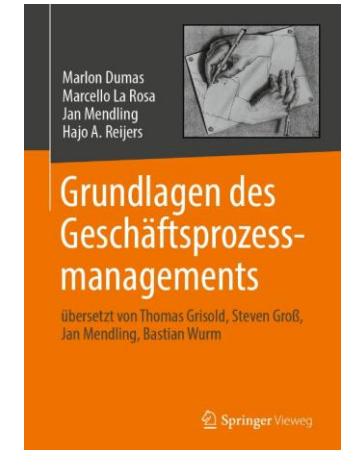
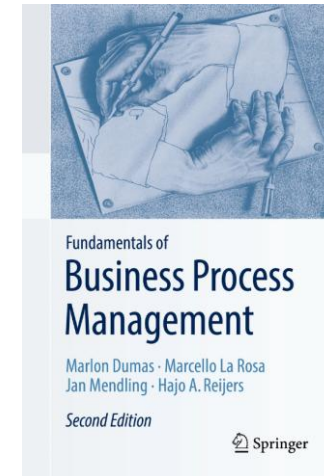
- Understand process monitoring with process dashboards
- Understand process discovery
- Understand process conformance checking
- Understand variants analysis and process performance mining

Recap Questions

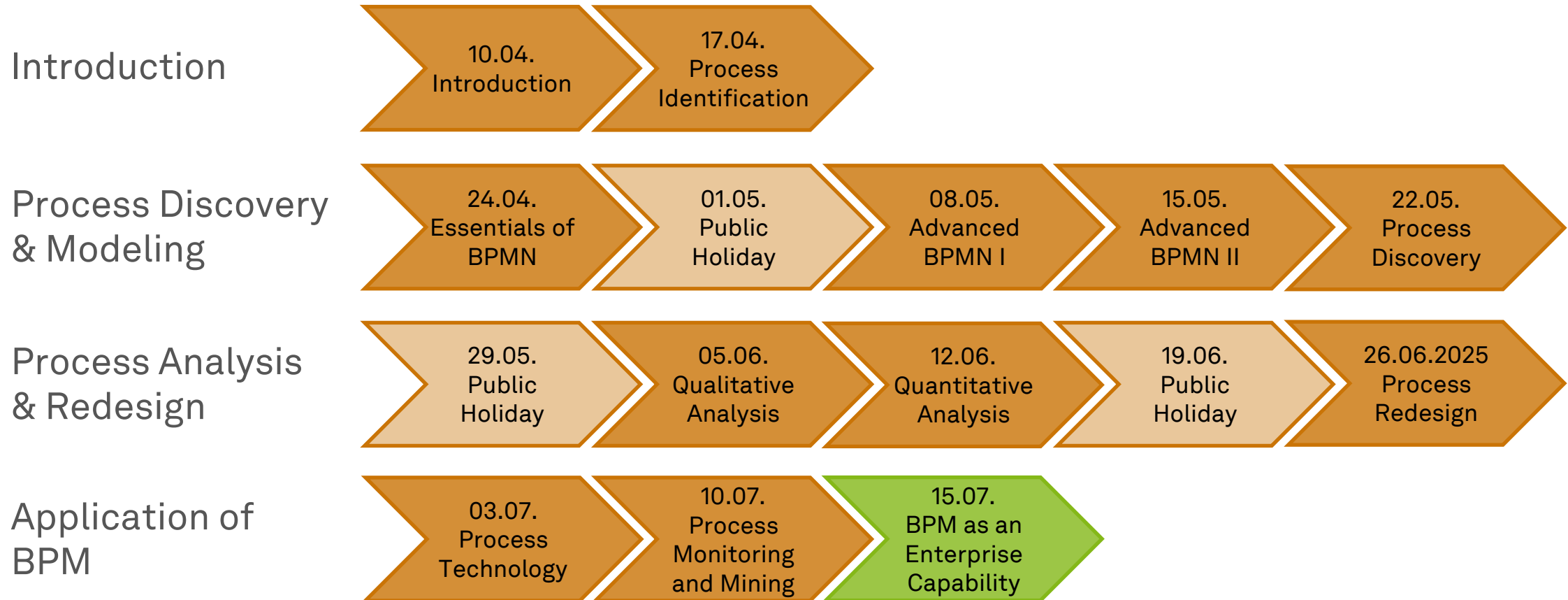
- What is the difference between offline and online process monitoring?
Which kinds of dashboards are best suited for which mode?
- How are the four distinct techniques of process mining related and how can they work together to optimize processes?

Literature

- **Section 11: Process Monitoring**
- **Section 2: Process Mining: The Missing Link**
- **Section 6: Process Discovery: An Introduction**
- **Section 8: Conformance Checking**



Roadmap



Any Questions about Process Monitoring and Process Mining?





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